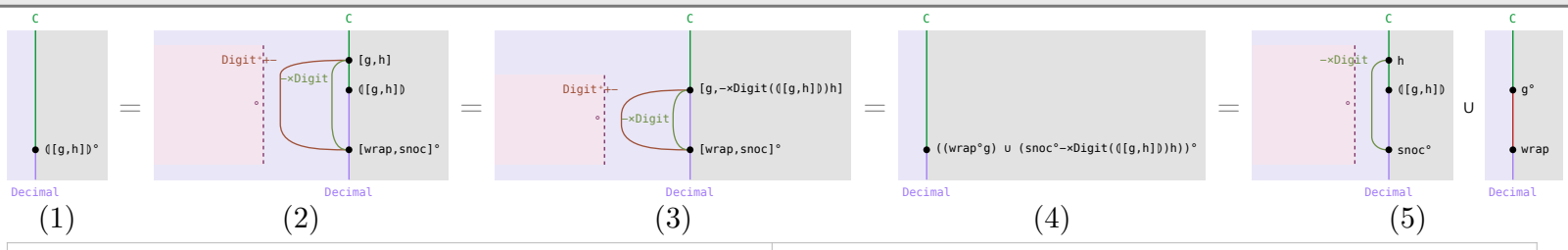
Companion to Bird & de Moor, Algebra of Programming

§6. Recursive Programs

§6.1. Digits of a number

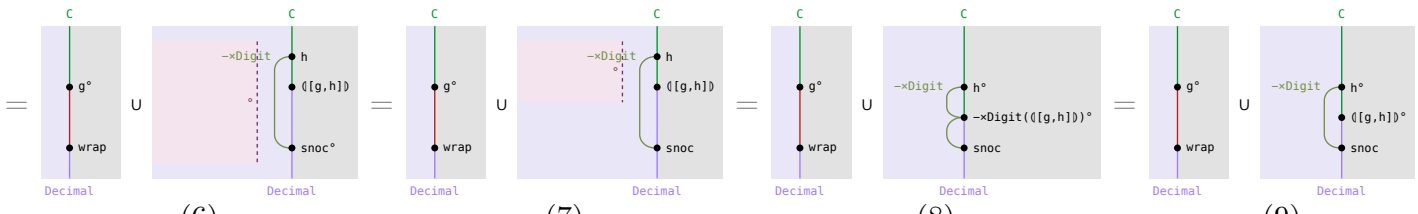
`⦇[g,h]⦈° = (g° wrap) ∪ (h° −×Digit(⦇[g,h]⦈°) snoc)` (6.1a)

the converse of a fold over decimals undoes the algebra `[g,h]` and then either `wrap`, or the fold's converse on the front and `snoc`



(1) (2) (3) (4) (5)

1	`⦇[g,h]⦈°`	`⦇φ⦈ = [wrap,snoc]° Digit⁺+−(−×Digit(⦇φ⦈)) φ`
2	`= ([wrap,snoc]° Digit⁺+−(−×Digit(⦇[g,h]⦈))) [g,h])°`	`(R + S) [P,Q] = [R P,S Q]`
3	`= ([wrap,snoc]° [g,−×Digit(⦇[g,h]⦈) h])°`	`[U,V]° [R,S] = (U° R) ∪ (V° S)`
4	`= ((wrap° g) ∪ (snoc° −×Digit(⦇[g,h]⦈) h))°`	`(R ∪ S)° = S° ∪ R°`
5	`= (snoc° −×Digit(⦇[g,h]⦈) h)° ∪ (wrap° g)°`	`R ∪ S = S ∪ R`



(6) (7) (8) (9)

6	`= (wrap° g)° ∪ (snoc° −×Digit(⦇[g,h]⦈) h)°`	`(R S)° = S° R°`
7	`= (g° wrap) ∪ ((−×Digit(⦇[g,h]⦈) h)° snoc)`	`(R h)° = h° R°`
8	`= (g° wrap) ∪ (h° −×Digit(⦇[g,h]⦈)° snoc)`	`−×Digit(R)° = −×Digit(R°)`
9	`= (g° wrap) ∪ (h° −×Digit(⦇[g,h]⦈°) snoc)`	

`val° = (embed° wrap) ∪ (op° −×Digit(val°) snoc)`

a number is read back into digits either as one nonzero digit, or by splitting off its last digit and reading back the rest

(6.1b)

1	`op°(m,p) ⇔ π₁(p) = m/10 ∧ π₂(p) = m % 10` `op°` splits a number into its quotient and remainder by 10
2	`(∃p. 0 < π₁(p) ∧ op°(m,p)) ⇔ 10 ≤ m` `op°` gives a pair with a nonzero first component exactly at the numbers with two or more digits
3	`(∃d. embed°(m,d)) ⇔ 0 < m ∧ m < 10` `embed°` gives a digit exactly at the one-digit numbers

§6.2. Least fixed points

(6.2a)

`Monotonic(φ) ≜ ∀X. ∀Y. X ⊑ Y ⇒ φ(X) ⊑ φ(Y)` — φ maps the hom-set `A↛B` into itself with `X⊑Y⇒φ(X)⊑φ(Y)`.

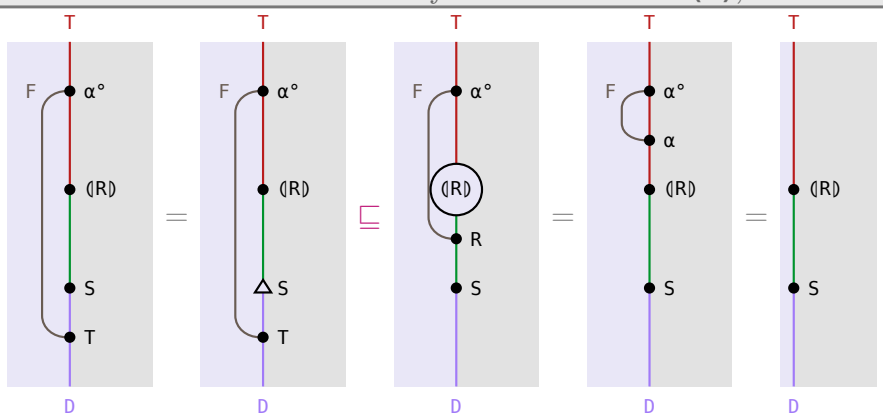
`(μX : φ(X))` is the least `X : A↛B` with `φ(X)⊑X`.

(6.2b)

	the law	what it says
1	`φ(Y) ⊑ Y ⇒ (μX : φ(X)) ⊑ Y`	to bound `(μX : φ(X))` above, exhibit one `Y` the body does not grow past — the half §Section 6.3. and every chapter after it uses
2	`Monotonic(φ) ⇒ φ(μX : φ(X)) = (μX : φ(X))`	**Knaster–Tarski**: the least solution of `φ(X)⊑X` already solves `φ(X)=X`, so the least prefix point and the least fixed point are one relation
3	`α° F(X) R ⊑ X ⇒ (R) ⊑ X` (6.2)	a relation `X` that contains one unfolding of the fold's recursion at `X` contains the fold
4	`X ⊑ α° F(X) R ⇒ X ⊑ (R)` (6.3)	a relation `X` contained in one unfolding of the fold's recursion at `X` is contained in the fold

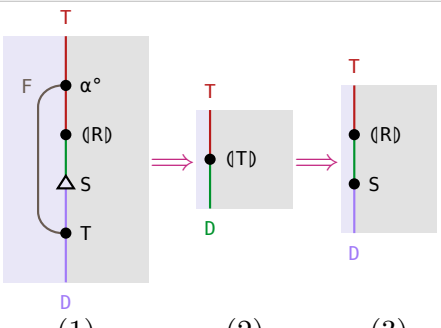
`F(S) T ⊑ R S`
`⇒ (T) ⊑ (R) S` (6.2c)

if **S** followed by **R** absorbs **T** after **F(S)**, the fold of **T** is below the fold of **R** followed by **S**



(1) (2) (3) (4) (5)

1	`α° F((R) S) T`	
2	`= α° F((R)) F(S) T`	`F(S) T ⊑ R S`
3	`⊑ α° F((R)) R S`	`α (R) = F((R)) R`
4	`= α° α (R) S`	`α° α = 𝟙`
5	`= (R) S`	



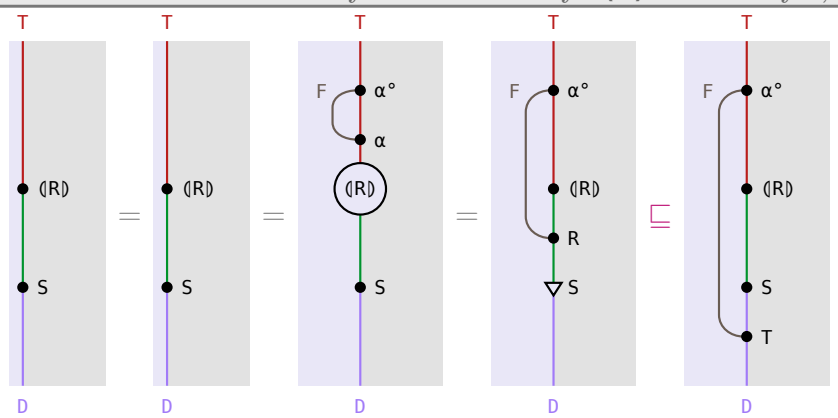
(1) (2) (3)

1	`F(S) T ⊑ R S`	`F(S) T ⊑ R S` `⇒ α° F((R) S) T ⊑ (R) S`
2	`⟹ α° F((R) S) T ⊑ (R) S`	`α° F(X) T ⊑ X` `⇒ (T) ⊑ X`
3	`⟹ (T) ⊑ (R) S`	

R S ⊑ F(S) T (6.2d)

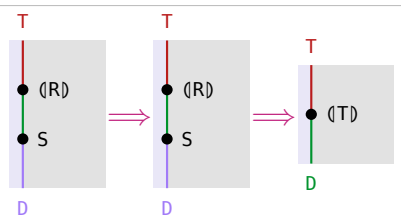
⇒ (R) S ⊑ (T)

if **R** followed by **S** is absorbed by **F(S)** followed by **T**, the fold of **R** followed by **S** is below the fold of **T**



(1) (2) (3) (4) (5)

1	(R) S	
2	= (R) S	α° α = 𝟙
3	= α° α (R) S	α (R) = F((R)) R
4	= α° F((R)) R S	R S ⊑ F(S) T
5	⊑ α° F((R) S) T	



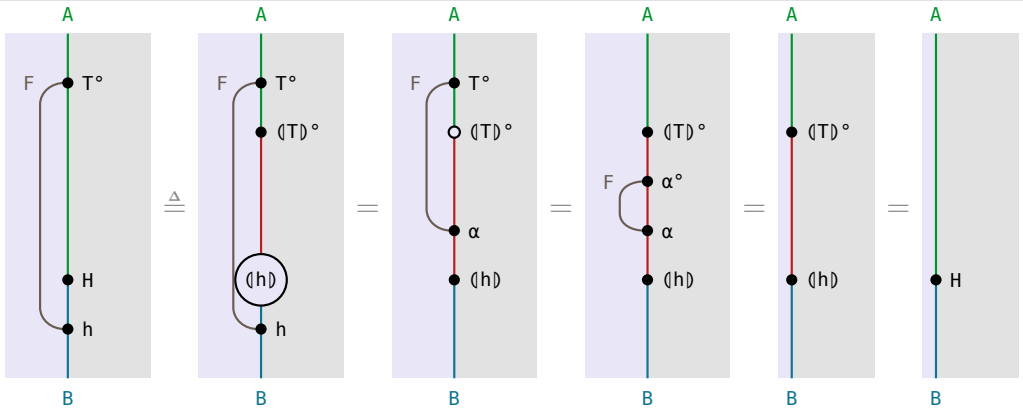
(1) (2) (3)

1	R S ⊑ F(S) T	R S ⊑ F(S) T ⇒ (R) S ⊑ α° F((R) S) T
2	⟹ (R) S ⊑ α° F((R) S) T	X ⊑ α° F(X) T ⇒ X ⊑ (T)
3	⟹ (R) S ⊑ (T)	

§6.3. Hylomorphisms

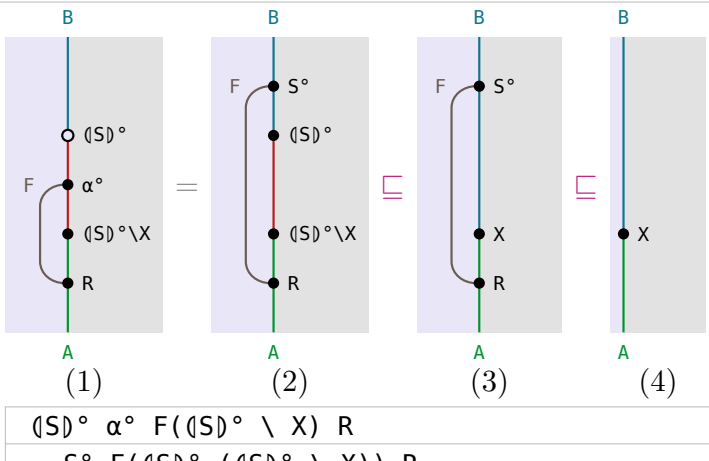
⦇S⦈° ⦇R⦈ = (μX : S° F(X) R) (6.3a)

hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation



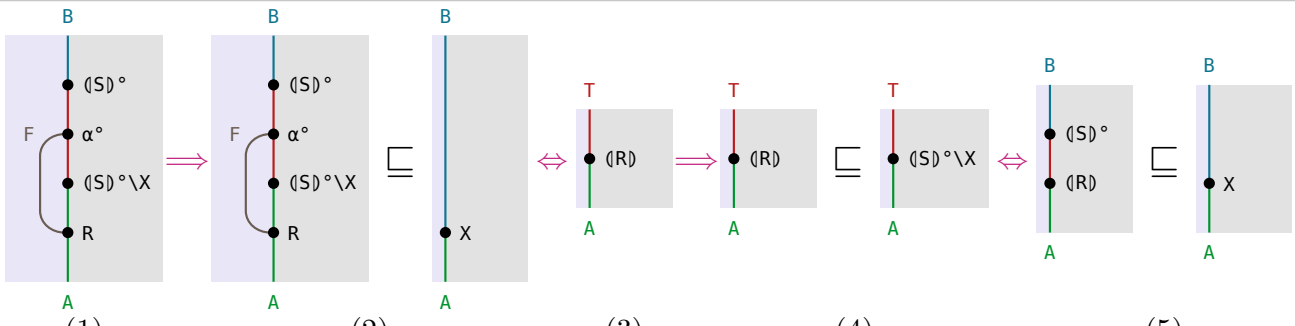
(1) (2) (3) (4) (5) (6)

1	T° F(H) h	H ≜ ⦇T⦈° ⦇h⦈
2	≜ T° F(⦇T⦈° ⦇h⦈) h	α ⦇h⦈ = F(⦇h⦈) h
3	= T° F(⦇T⦈°) α ⦇h⦈	⦇T⦈° α° = T° F(⦇T⦈°)
4	= ⦇T⦈° α° α ⦇h⦈	α° α = 𝟙
5	= ⦇T⦈° ⦇h⦈	H = ⦇T⦈° ⦇h⦈
6	= H	



(1) (2) (3) (4)

1	⦇S⦈° α° F(⦇S⦈° \ X) R	⦇S⦈° α° = S° F(⦇S⦈°)
2	= S° F(⦇S⦈° (⦇S⦈° \ X)) R	S (S \ X) ⊑ X
3	⊑ S° F(X) R	S° F(X) R ⊑ X
4	⊑ X	

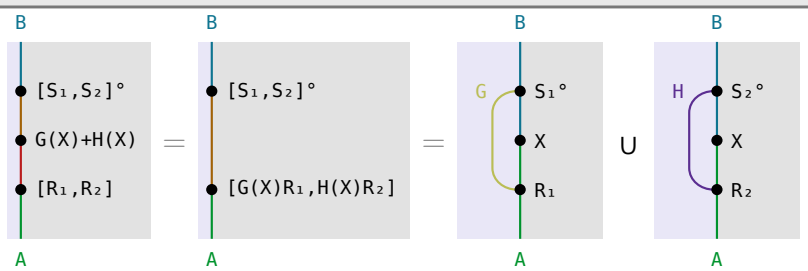


(1) (2) (3) (4) (5)

1	S° F(X) R ⊑ X	S° F(X) R ⊑ X ⇒ ⦇S⦈° α° F(⦇S⦈° \ X) R ⊑ X
2	⟹ ⦇S⦈° α° F(⦇S⦈° \ X) R ⊑ X	X ⊑ S \ R ⇔ S X ⊑ R
3	⟺ α° F(⦇S⦈° \ X) R ⊑ ⦇S⦈° \ X	α° F(X) R ⊑ X ⇒ ⦇R⦈ ⊑ X
4	⟹ ⦇R⦈ ⊑ ⦇S⦈° \ X	X ⊑ S \ R ⇔ S X ⊑ R
5	⟺ ⦇S⦈° ⦇R⦈ ⊑ X	

⦇[S₁,S₂]⦈° ⦇[R₁,R₂]⦈ = (μX : (S₁° G(X) R₁) ∪ (S₂° H(X) R₂)) (6.3b)

when both algebras are case splits over the same sum, the recursion runs each case on its own and unites the results



(1) (2) (3)

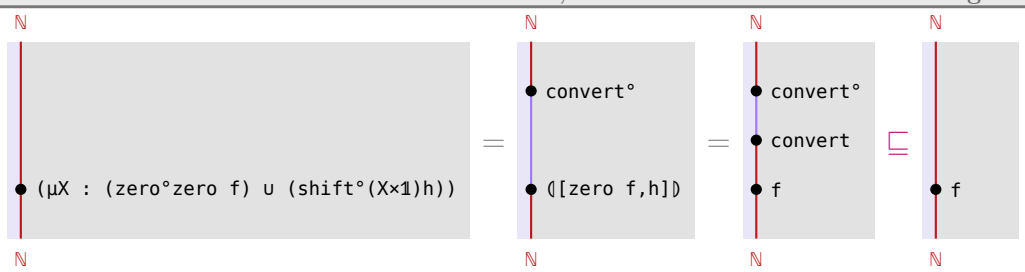
1	`[S₁,S₂]° (G(X) + H(X)) [R₁,R₂]`	`(R + S) [R₁,R₂] = [R R₁,S R₂]`
2	`= [S₁,S₂]° [G(X) R₁,H(X) R₂]`	`[S₁,S₂]° [R,S] = (S₁° R) ∪ (S₂° S)`
3	`= (S₁° G(X) R₁) ∪ (S₂° H(X) R₂)`	

§6.4. Fast exponentiation and modulus computation

shift f = (f × 𝟙) h (6.4a)

⇒ (μX : (zero° zero f) ∪ (shift° (X × 𝟙) h)) ⊑ f

once **f** turns **shift** into **h**, the recursion that halves the argument at each step computes **f**



1	(μX : (zero° zero f) ∪ (shift° (X × 𝟙) h))	convert° ⦇[g,h]⦈ = (μX : (zero° g) ∪ (shift° (X × 𝟙) h))
2	= convert° ⦇[zero f,h]⦈	zero f = g; shift f = (f × 𝟙) h ⇒ convert f = ⦇[g,h]⦈
3	= convert° convert f	Simple(convert)
4	⊑ f	

(μX : (zero° one) ∪ (shift° (X × 𝟙) op(a))) ⊑ exp(a) (6.4b)

the recursion that halves the exponent at each step computes **a** to the power **b**

`zero exp(a) = one`

`a` to the power `0` is `1`

`shift exp(a) = (exp(a) × 𝟙) op(a)`

`a` to the power `2n+d` is `op(a)` of `a` to the power `n` and `d`

(μX : (zero° zero) ∪ (shift° (X × 𝟙) op(b))) ⊑ mod(b) (6.4c)

the recursion that halves **a** at each step computes the remainder of **a** by **b**

`zero mod(b) = zero`

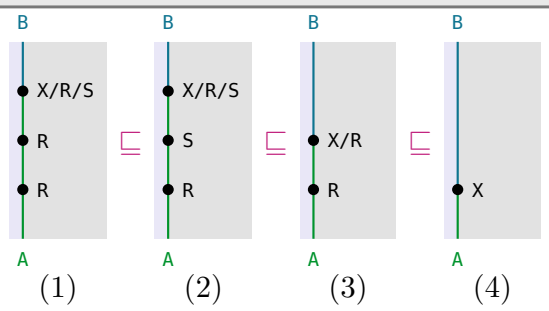
`0 mod b` is `0`

`shift mod(b) = (mod(b) × 𝟙) op(b)`

`(2a+d) mod b` is `op(b)` of `a mod b` and `d`

§6.5. Unique fixed points

$$R^2 \sqsubseteq S\ R \quad (6.5a)$$
$$\Rightarrow (X\ /\ R\ /\ S)\ R^2 \sqsubseteq X$$
((X/R)/S) followed by RR stays below X when RR⊑SR
inductive(S); $R^2 \sqsubseteq S\ R$
⇒ inductive(R)
R is inductive when RR⊑SR for an inductive S



1	(X / R / S) R^2	$R^2 \sqsubseteq S\ R$
2	$\sqsubseteq$ (X / R / S) S R	(R / S) S ⊑ R
3	$\sqsubseteq$ (X / R) R	(X / R) R ⊑ X
4	$\sqsubseteq$ X	

§6.5.1. Membership

member = ∅ (6.5.1a)
a constant relator records no elements, so its membership is empty
member = [member,member]
a member of F+G is a member of whichever summand the value is in
member = (π_1 member) ∪ (π_2 member)
a member of F×G is a member of either component

§6.5.2. Consequences

inductive(S member); X = S F(X) R; Y = S F(Y) R (6.5.2a)
⇒ X = Y
when S member is inductive, two solutions X, Y of X=SF(X)R are equal

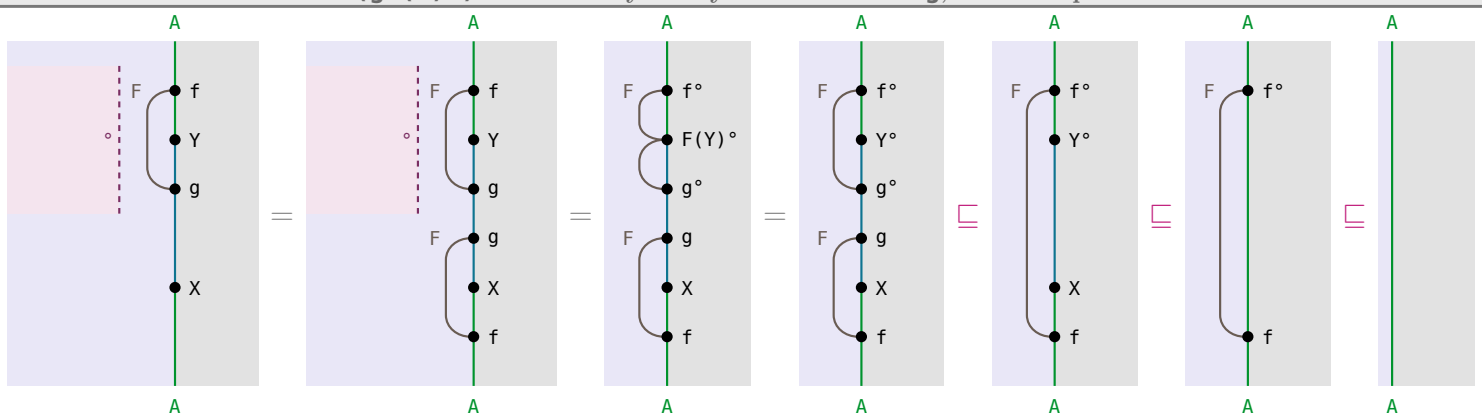
inductive(S member); Entire(S); Entire(R); S F(X) R ⊑ X (6.5.2b)
⇒ Entire(X)
when S member is inductive and S, R are entire, a pre-fixed point of X↦SF(X)R is entire

inductive(g member); Map(g); Map(f); X = g F(X) f
⇒ Map(X) (6.5.2c)

when **g member** is inductive and **f**, **g** are maps, the solution of **X=gF(X)f** is a map

X = g F(X) f; Simple(g); Simple(f); Y° X ⊑ 𝟙
⇒ (g F(Y) f)° X ⊑ 𝟙

(gF(Y)f)° followed by **X** stays below **𝟙** when **g**, **f** are simple and **Y°X⊑𝟙**

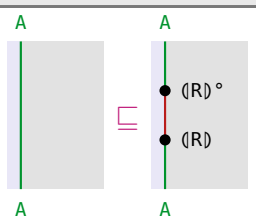


(1) (2) (3) (4) (5) (6) (7)

1	(g F(Y) f)° X	X = g F(X) f
2	= (g F(Y) f)° g F(X) f	(R S)° = S° R°
3	= f° F(Y)° g° g F(X) f	PreservesRecip(F)
4	= f° F(Y°) g° g F(X) f	Simple(g)
5	⊑ f° F(Y° X) f	Y° X ⊑ 𝟙
6	⊑ f° f	Simple(f)
7	⊑ 𝟙	

inductive(R° member); 𝟙 ⊑ R° R
⇒ 𝟙 ⊑ (R)° (R) (6.5.2d)

when **R°member** is inductive and **R** is surjective, the fold **(R)** is surjective

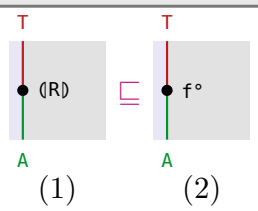


(1) (2)

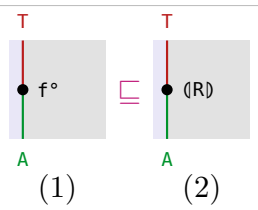
1	𝟙	inductive(S member); Entire(S); Entire(R); S F(X) R ⊑ X ⇒ Entire(X) at S=R°
2	⊑ (R)° (R)	

Map(f); R f ⊑ F(f) α; 𝟙 ⊑ R° R
⇒ f° = ⦇R⦈ (6.5.2e)

a map f with Rf⊑F(f)α, for a surjective R, has the fold ⦇R⦈ as its converse



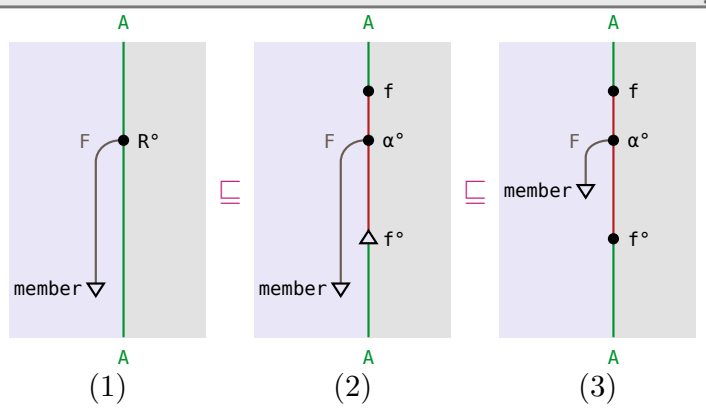
1	⦇R⦈	shunting f, fusion, Rf⊑F(f)α
2	⊑ f°	



1	f°	inductive(R° member); 𝟙 ⊑ R° R ⇒ 𝟙 ⊑ ⦇R⦈° ⦇R⦈ and ⦇R⦈f⊑𝟙
2	⊑ ⦇R⦈	

Map(f); R f ⊑ F(f) α
⇒ R° member ⊑ f α° member f° (6.5.2f)

R°member is below α°member conjugated by f, so it is inductive when α°member is



1	R° member	Map(f); R f ⊑ F(f) α ⇒ R° ⊑ f α° F(f°)
2	⊑ f α° F(f°) member	F(R) member ⊑ member R
3	⊑ f α° member f°	

§6.6. Sorting by selection

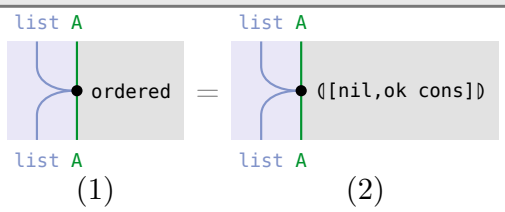
§6.6.1. The function sort

(6.6.1a)

	definition	type	meaning
1	`ordered[] = true, ordered([a]#xs) = (∀x. x ∈ xs ⇒ R(a,x)) ∧ ordered(xs)`	`[A]→ [A]`	every element is R-below each element after it
2	`(a,xs) ok q ⇔ (a,xs) = q ∧ (∀x. x ∈ xs ⇒ R(a,x))`	`A×[A]→ A×[A]`	the coreflexive passing `(a, x)` when `a` is R-below every element of `x`

ordered = (|[nil,ok cons]|) (6.6.1b)

a list is sorted exactly when the fold that rebuilds it passes **ok** at every **cons**, i.e. each head is **R**-below every element after it

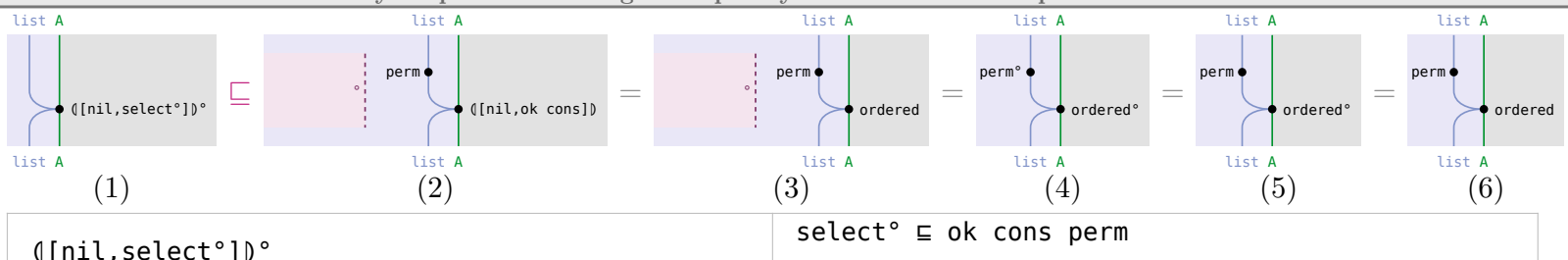


1	`ordered`	fold uniqueness
2	`= (|[nil,ok cons]|)`	

§6.6.2. Selection sort

select° ⊑ ok cons perm
⇒ (|[nil,select°]|)° ⊑ perm ordered (6.6.2a)

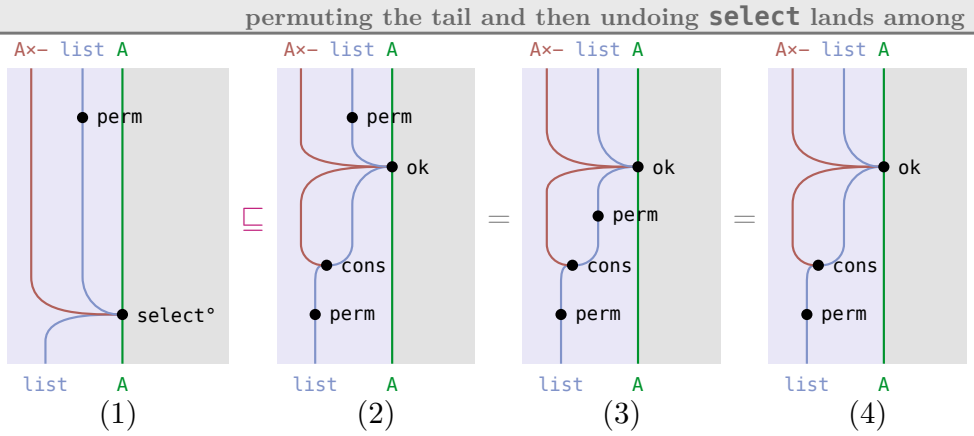
every output of unfolding the input by **select** is a sorted permutation of it



1	`(|[nil,select°]|)°`	`select° ⊑ ok cons perm` `⇒ (|[nil,select°]|)° ⊑ ((|[nil,ok cons]|) perm)°`
2	`⊑ ((|[nil,ok cons]|) perm)°`	(6.6.1b) `ordered = (|[nil,ok cons]|)`
3	`= (ordered perm)°`	`(ordered perm)° = perm° ordered°`
4	`= perm° ordered°`	`perm° = perm`
5	`= perm ordered°`	`ordered° = ordered`
6	`= perm ordered`	

select° ⊑ ok cons perm
⇒ (𝟙 × perm) select° ⊑ ok cons perm (6.6.2b)

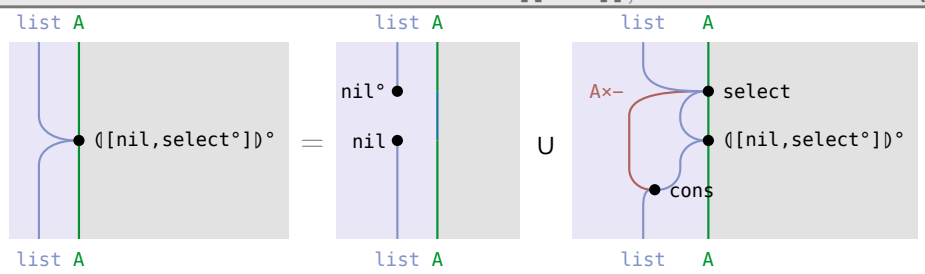
permuting the tail and then undoing **select** lands among the **ok** conses of a permutation



1	`(𝟙 × perm) select°`	`select° ⊑ ok cons perm`
2	`⊑ (𝟙 × perm) ok cons perm`	`ok (𝟙 × perm) = (𝟙 × perm) ok`
3	`= ok (𝟙 × perm) cons perm`	`cons perm = (𝟙 × perm) cons perm`
4	`= ok cons perm`	

`◖[nil,select°]◗° = (nil° nil) ∪ (select (𝟙 × ◖[nil,select°]◗°) cons)` (6.6.2c)

selection sort returns `[]` on `[]`, and otherwise selects `(a,y)`, sorts `y` and conses `a` back on

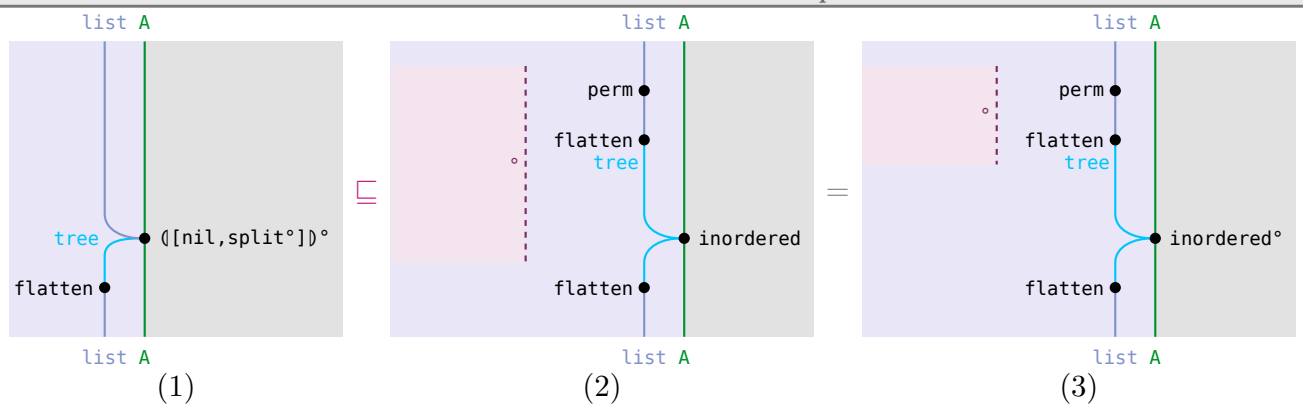


1	`◖[nil,select°]◗°`	unfolding the converse of a fold
2	`= (nil° nil) ∪ (select (𝟙 × ◖[nil,select°]◗°) cons)`	

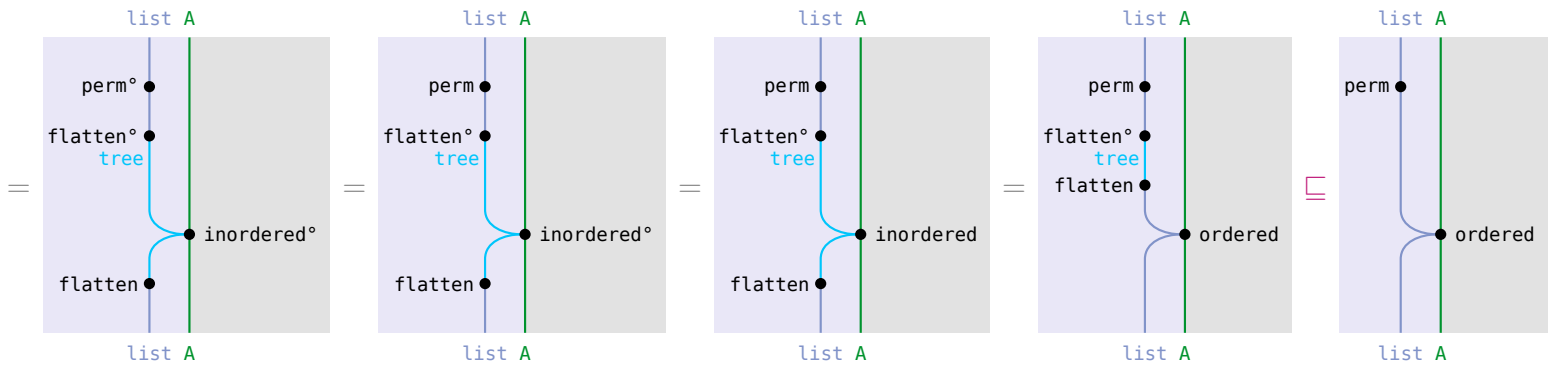
§6.6.3. Quicksort

`(∀a. ∀b. ∀c. R(a,b) ⇒ R(b,c) ⇒ R(a,c)); split° ⊑ check' join perm ⇒ ◖[nil,split°]◗° flatten ⊑ perm ordered` (6.6.3a)

every output of unfolding the input by **`split`** into a tree and flattening that tree is a sorted permutation of the input



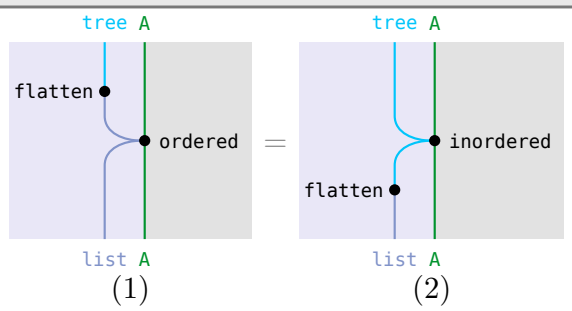
1	`◖[nil,split°]◗° flatten`	`split° ⊑ check' join perm ⇒ ◖[nil,split°]◗° flatten ⊑ (inordered flatten perm)° flatten`
2	`⊑ (inordered flatten perm)° flatten`	`(inordered S)° = S° inordered°`
3	`= (flatten perm)° inordered° flatten`	`(flatten perm)° = perm° flatten°`



4	`= perm° flatten° inordered° flatten`	`perm° = perm`
5	`= perm flatten° inordered° flatten`	`inordered° = inordered`
6	`= perm flatten° inordered flatten`	(6.6.3b) `(∀a. ∀b. ∀c. R(a,b) ⇒ R(b,c) ⇒ R(a,c)) ⇒ flatten ordered = inordered flatten`
7	`= perm flatten° flatten ordered`	`flatten° flatten ⊑ 𝟙`
8	`⊑ perm ordered`	

(∀a. ∀b. ∀c. R(a,b) ⇒ R(b,c) ⇒ R(a,c))
⇒ flatten ordered = inordered flatten (6.6.3b)

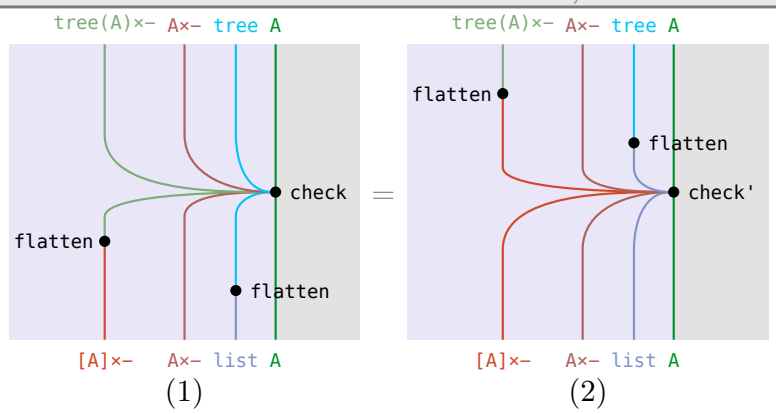
flattening a tree and then testing the list for order is the same as testing the tree with **inordered** and then flattening



1	`flatten ordered`	R transitive: a flattened tree is ordered iff every node passes `check`
2	`= inordered flatten`	

check (flatten × (𝟙 × flatten)) = (flatten × (𝟙 × flatten)) check' (6.6.3c)

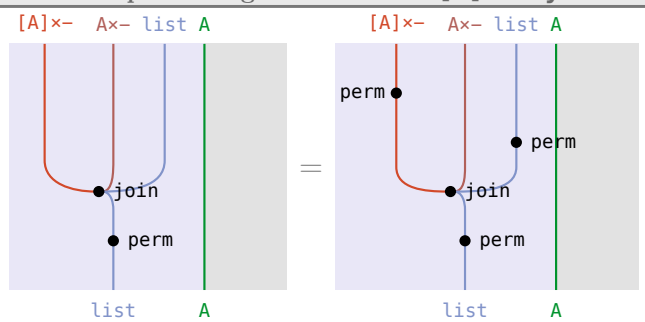
testing a fork with **check** and then flattening both subtrees equals flattening both subtrees and then testing with **check'**, since **intree** of a tree is **inlist** of its flattening



1	`check (flatten × (𝟙 × flatten))`	`b intree x ⇔ b inlist flatten(x)`
2	`= (flatten × (𝟙 × flatten)) check'`	

join perm = (perm × (𝟙 × perm)) join perm (6.6.3d)

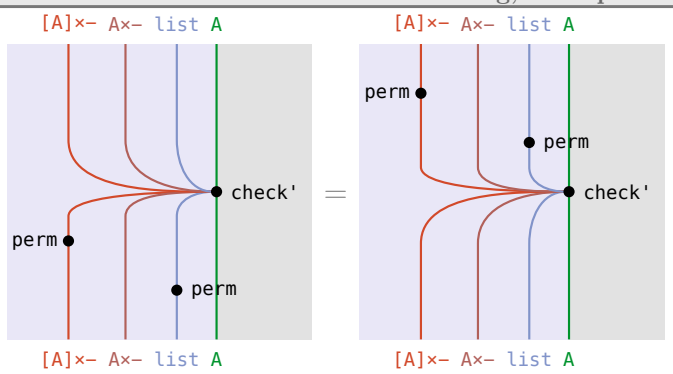
permuting the list **x ++ [a] ++ y** is the same as first permuting **x** and **y** and then permuting the joined list



1	`join perm`	`perm` is transitive and closed under `++`
2	`= (perm × (𝟙 × perm)) join perm`	

check' (perm × (𝟙 × perm)) = (perm × (𝟙 × perm)) check' (6.6.3e)

testing a fork of lists with **check'** and then permuting both parts equals permuting both parts and then testing, since permuting a list leaves its members unchanged

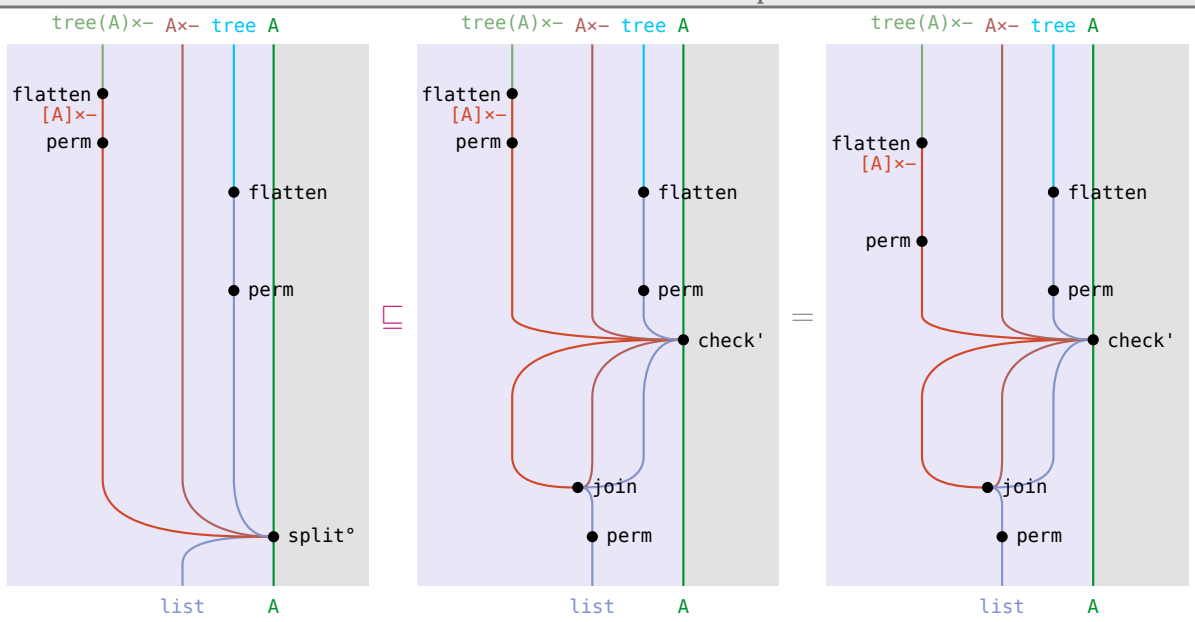



```
1  check' (perm × (𝟙 × perm))          b inlist x ⇔ b inlist perm(x)
2  = (perm × (𝟙 × perm)) check'
```

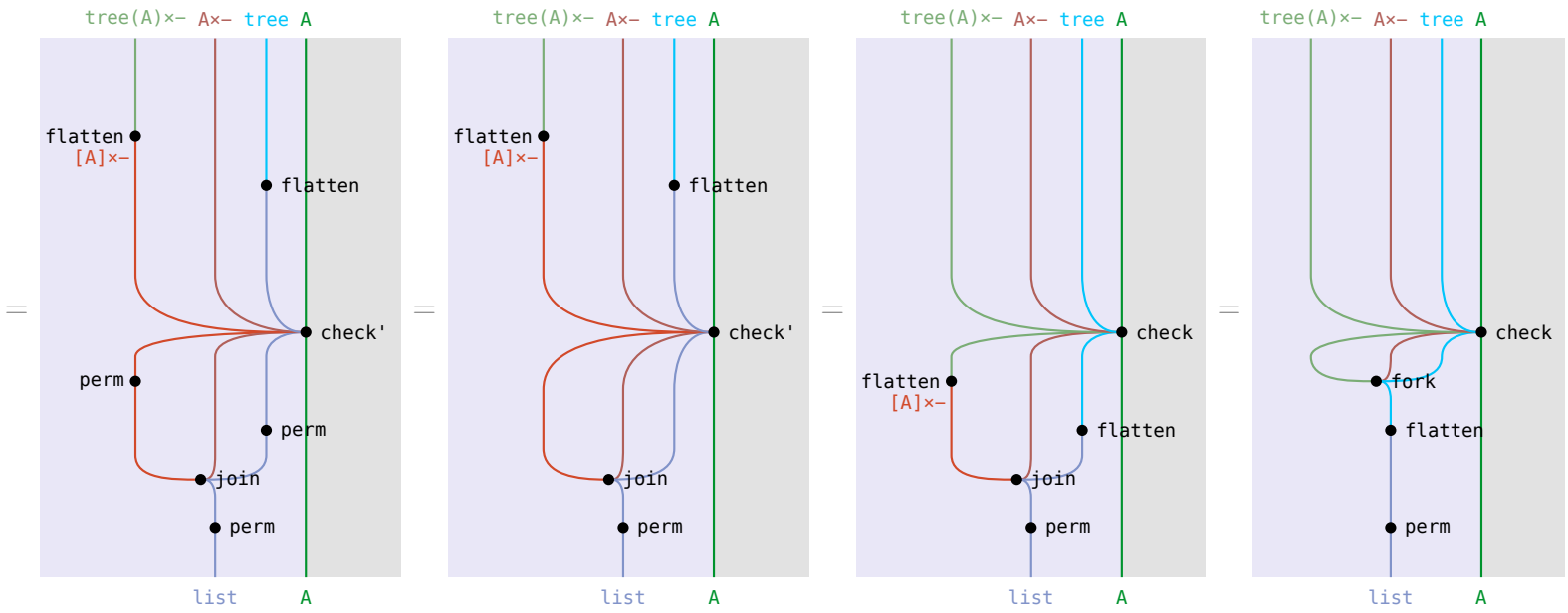
split° ⊑ check' join perm **(6.6.3f)**

⇒ ((flatten perm) × (𝟙 × (flatten perm))) split° ⊑ check fork flatten perm

undoing **split** and then flattening and permuting both parts lands among the **check**ed forks whose flattening is permuted



1	((flatten perm) × (𝟙 × (flatten perm))) split°	split° ⊑ check' join perm
2	⊑ ((flatten perm) × (𝟙 × (flatten perm))) check' join perm	
3	= (flatten × (𝟙 × flatten)) (perm × (𝟙 × perm)) check' join perm	(6.6.3e) check' (perm × (𝟙 × perm)) = (perm × (𝟙 × perm)) check'

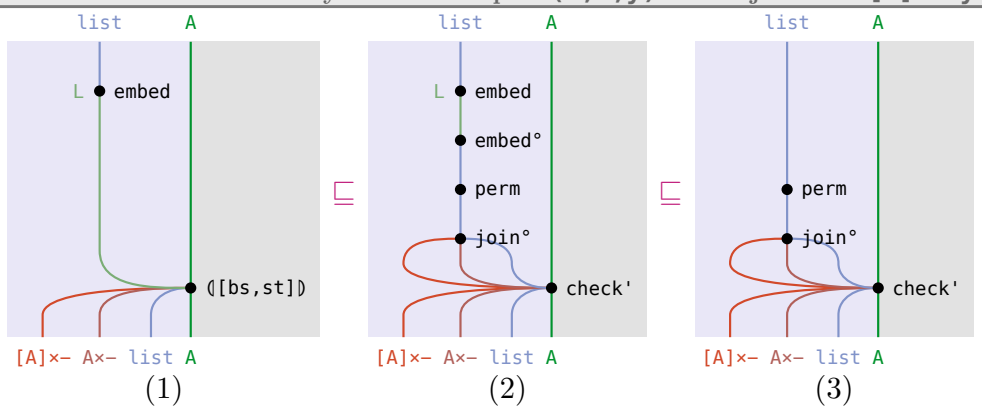


4	= (flatten × (𝟙 × flatten)) check' (perm × (𝟙 × perm)) join perm	(6.6.3d) join perm = (perm × (𝟙 × perm)) join perm
5	= (flatten × (𝟙 × flatten)) check' join perm	(6.6.3c) check (flatten × (𝟙 × flatten)) = (flatten × (𝟙 × flatten)) check'
6	= check (flatten × (𝟙 × flatten)) join perm	fork flatten = (flatten × (𝟙 × flatten)) join
7	= check fork flatten perm	

`bs ⊑ τ perm join° check'; (𝟙 × (perm join° check')) st ⊑ cons perm join° check'` (6.6.3g)

`⇒ embed (|[bs,st]|) ⊑ perm join° check'`

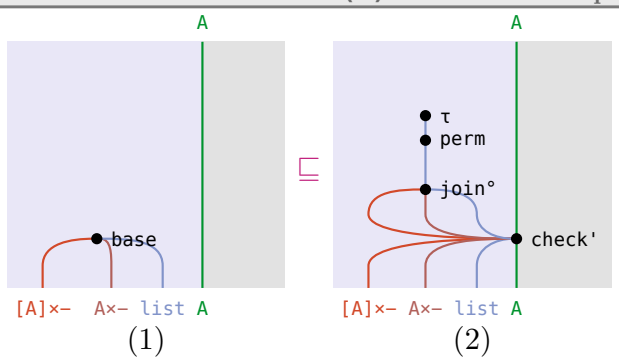
turning a list into a non-empty list and folding it with any `[bs, st]` meeting the two conditions below gives only **check**ed triples `(x,a,y)` whose join `x ⧺ [a] ⧺ y` is a permutation of the list



1	`embed (|[bs,st]|)`	`bs ⊑ τ perm join° check'; (𝟙 × (perm join° check')) st ⊑ cons perm join° check' ⇒ embed (|[bs,st]|) ⊑ embed embed° perm join° check'`
2	`⊑ embed embed° perm join° check'`	`embed embed° perm join° check' ⊑ perm join° check'`
3	`⊑ perm join° check'`	

`base ⊑ τ perm join° check'` (6.6.3h)

`base(a)` is a **check**ed triple whose join is a permutation of the one-element list `[a]`

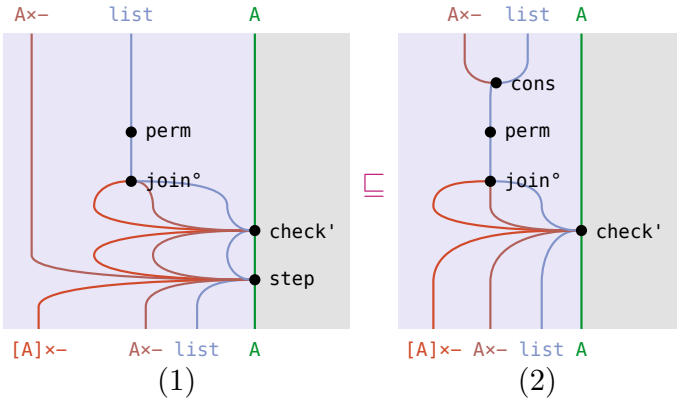


1	`base`	`[] ⧺ [a] ⧺ [] = [a]`, and `check'` holds on empty lists
2	`⊑ τ perm join° check'`	

`connected(R)` (6.6.3i)

`⇒ (𝟙 × (perm join° check')) step ⊑ cons perm join° check'`

if **`(x,b,y)`** is a **check**ed triple for a permutation of **`l`**, then **`step(a,(x,b,y))`** is a **check**ed triple for a permutation of **`[a] ⧺ l`**

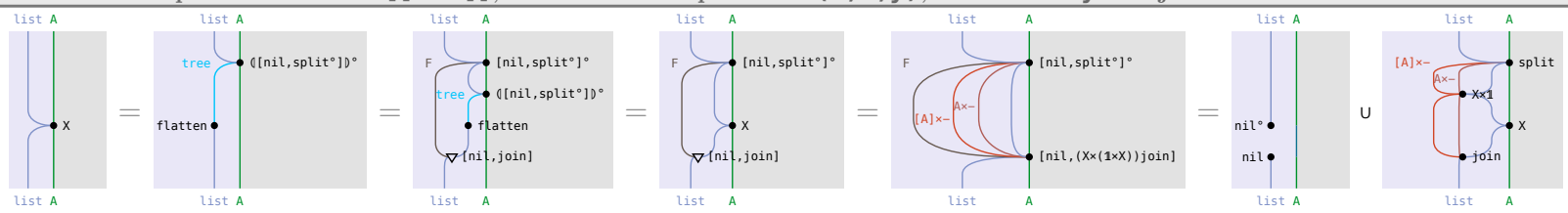


1	`(𝟙 × (perm join° check')) step`	`aRb` puts `a` in front of `x`; otherwise `bRa` and `a` goes in front of `y`
2	`⊑ cons perm join° check'`	

X = ◖[nil,split°]◗° flatten (6.6.3j)

⇒ X = (nil° nil) ∪ (split (X × (𝟙 × X)) join)

quicksort returns `[]` on `[]`, and otherwise splits into `(x,a,y)`, sorts `x` and `y` and joins them around `a`



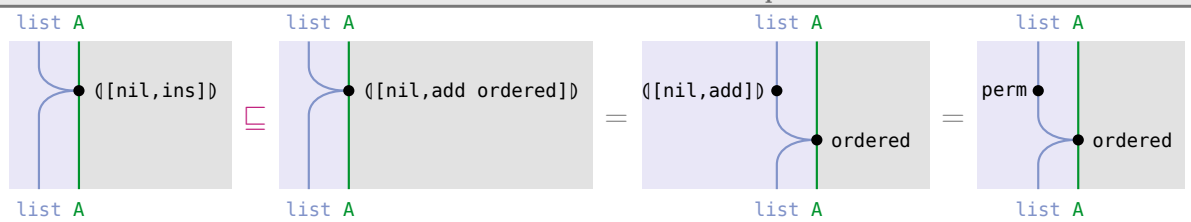
(1) (2) (3) (4) (5) (6)

1	X	X = ◖[nil,split°]◗° flatten
2	= ◖[nil,split°]◗° flatten	T° F(H) h = H
3	= [nil,split°]° F(◖[nil,split°]◗° flatten) [nil,join]	X = ◖[nil,split°]◗° flatten
4	= [nil,split°]° F(X) [nil,join]	F(X) [nil,join] = [nil,(X × (𝟙 × X)) join]
5	= [nil,split°]° [nil,(X × (𝟙 × X)) join]	[nil,V]° [nil,S] = (nil° nil) ∪ (V° S)
6	= (nil° nil) ∪ (split (X × (𝟙 × X)) join)	

(𝟙 × ordered) ins ⊑ add ordered (6.6.3k)

⇒ ◖[nil,ins]◗ ⊑ perm ordered

for any `ins` that, on an ordered list, returns only ordered results of `add`, folding with `[nil, ins]` gives only sorted permutations



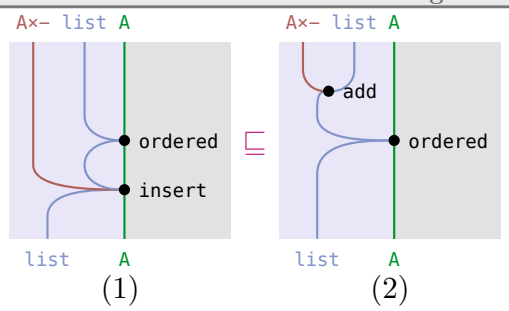
(1) (2) (3) (4)

1	◖[nil,ins]◗	(𝟙 × ordered) ins ⊑ add ordered ⇒ ◖[nil,ins]◗ ⊑ ◖[nil,add ordered]◗
2	⊑ ◖[nil,add ordered]◗	◖[nil,add]◗ ordered = ◖[nil,add ordered]◗
3	= ◖[nil,add]◗ ordered	perm = ◖[nil,add]◗
4	= perm ordered	

connected(R); (∀a. ∀b. ∀c. R(a,b) ⇒ R(b,c) ⇒ R(a,c)) (6.6.3l)

⇒ (𝟙 × ordered) insert ⊑ add ordered

inserting `a` into an ordered list puts `a` somewhere in it and keeps it ordered



(1) (2)

1	(𝟙 × ordered) insert	insert splices a in, and R transitive and connected
2	⊑ add ordered	

§6.7. Closure

R* = (μX : 𝟙 ∪ (X R)) (6.7a)

closing R by composing it on the left and closing it by composing on the right give the same relation R*

𝟙 ⊑ X; X² ⊑ X
⇒ R ⊑ X ⇔ R* ⊑ X

X contains R exactly when it contains R*

𝟙 ⊑ R* (6.7c)

R* is reflexive

𝟙	R ⊑ R ∪ S
⊑ 𝟙 ∪ (R R*)	𝟙 ∪ (R R*) = R*
= R*	

R ⊑ R* (6.7d)

R* contains R

R	
= R	(6.7c) 𝟙 ⊑ R*
⊑ R R*	R R* ⊑ R*
⊑ R*	

(𝟙 ∪ (R (R* / R*))) R* ⊑ R* (6.7e)

R*/R* is a prefixed point of X ↦ 𝟙∪RX, stated through division

R*² ⊑ R*

R* is transitive, because R*/R* is a prefixed point of X ↦ 𝟙∪RX

(𝟙 ∪ (R (R* / R*))) R*	(S ∪ T) R = (S R) ∪ (T R)
= (R*) ∪ (R (R* / R*) R*)	
= R* ∪ (R (R* / R*) R*)	(X / X) X ⊑ X
⊑ R* ∪ (R R*)	R R* ⊑ R*
⊑ R*	

𝟙 ⊑ X; X² ⊑ X; R ⊑ X (6.7f)
⇒ 𝟙 ∪ (R X) ⊑ X

a reflexive transitive X containing R is a prefixed point of Y ↦ 𝟙∪RY

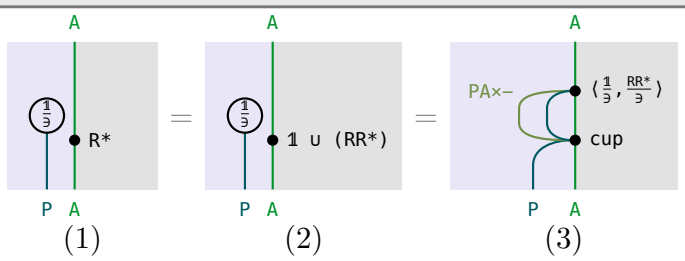
𝟙 ⊑ X; X² ⊑ X; R ⊑ X
⇒ R* ⊑ X

a reflexive transitive X containing R is a prefixed point of X ↦ 𝟙∪RX, so it contains R*

𝟙 ∪ (R X)	𝟙 ⊑ X
⊑ X ∪ (R X)	R ⊑ X
⊑ X ∪ (X²)	
⊑ X	

$\frac{R^*}{∋}$ = $\langle \frac{𝟙}{∋}, \frac{R\ R^*}{∋} \rangle$ cup (6.7g)

the set of R*-successors of a is a itself joined with the R*-successors of its R-successors



$\frac{R^*}{∋}$	𝟙 ∪ (R R*) = R*
= $\frac{𝟙 ∪ (R\ R^*)}{∋}$	$\frac{R ∪ S}{∋}$ = $\langle \frac{R}{∋}, \frac{S}{∋} \rangle$ cup
= $\langle \frac{𝟙}{∋}, \frac{R\ R^*}{∋} \rangle$ cup	

§6.7.1. Computing closure

R − ∅ = R (6.7.1a)

taking nothing away leaves R

R ∪ (S − R) = R ∪ S

beside R, only the part of S outside R adds anything

R − (S ∪ T) = R − S − T

taking away S and then T takes away S∪T

(R ∪ S) − T = (R − T) ∪ (S − T)

taking T away from a union takes it away from each part

monotonic(φ); monotonic(ψ) (6.7.1b)

⇒ (μX : φ(ψ(X))) = φ(μY : ψ(φ(Y)))

the least fixed point of φ after ψ is φ applied to the least fixed point of ψ after φ

S R* = (μX : S ∪ (X R)) (6.7.1c)

S followed by any number of R steps is the least X containing S and closed under a further R step

R* S = (μX : S ∪ (R X))

any number of R steps followed by S is the least X containing S and closed under an R step in front

θ(∅,S) = S R* (6.7.1d)

started with nothing found and S to explore, θ computes SR*

1	θ(∅,S)	θ(P,S) ≜ P ∪ (μX : S ∪ ((X R) − P))
2	≜ ∅ ∪ (μX : S ∪ ((X R) − ∅))	(6.7.1a) R − ∅ = R
3	= ∅ ∪ (μX : S ∪ (X R))	∅ ∪ R = R
4	= (μX : S ∪ (X R))	(6.7.1c) S R* = (μX : S ∪ (X R))
5	= S R*	

θ(P,∅) = P (6.7.1e)

with nothing left to explore, θ returns what it has found

1	θ(P,∅)	θ(P,Q) ≜ P ∪ (μX : Q ∪ ((X R) − P))
2	≜ P ∪ (μX : ∅ ∪ ((X R) − P))	(μX : ∅ ∪ ((X R) − P)) = ∅
3	= P ∪ ∅	P ∪ ∅ = P
4	= P	